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10. P_A^o = 10 torr , $P_{solution}$ = 9 torr

W_A = 20 g W_B = 1 g

M_A = 200 g mol^{-1} M_B = ?

$$\frac{\Delta P}{P_A^o} = \frac{W_B \times M_A}{M_B \times W_A}$$

$$\frac{10-9}{10} = \frac{1 \times 200}{M_B \times 20}$$

$$M_B = \frac{200}{20} \times 10 = 100 \text{ g mol}^{-1}$$

11. W_2 = 2.56 g

W_1 = 100 g

T = 319.692 K

K_b = 2.42 K Kg mol^{-1}

ΔT_b = (319.692 – 319.450) K

= 0.242 K

$$M_2 = \frac{K_b \times W_2 \times 100}{\Delta T_b \times W_1}$$

$$= \frac{2.42 \times 2.56 \times 1000}{0.242 \times 100}$$

$$M_2 = 256 \text{ g mol}^{-1}$$

Molecular mass of sulphur in solution
= 256 g mol^{-1}

atomic mass of one mole of sulphur atom = 32

No. of atoms in a molecule of sulphur

$$= \frac{256}{32} = 8$$

Hence molecular formula of sulphur is S_8.

12. W_2 = 2g W_1 = 75 g

ΔT_f = 0.2 K K_f = 5.12 K Kg mol^{-1}

M_2 = ?

$$M_2 = \frac{K_f \times W_2 \times 1000}{\Delta T_f \times W_1} = \frac{5.12 \times 2 \times 1000}{0.2 \times 75}$$

$$= 682.66 \text{ g mol}^{-1}$$

13.Osmotic pressure of urea solution (π_1)=CRT

$$= \frac{W_2}{M_2 V} RT$$

$$= \frac{6}{60 \times 1} \times RT$$

Osmotic pressure of glucose solution

$$(\pi_2) = \frac{W_2}{180 \times 1} \times RT$$

For isotonic solution,

$\pi_1 = \pi_2$

$$\frac{6}{60} RT = \frac{W_2}{180} RT$$

$$\Rightarrow W_2 = \frac{6}{60} \times 180$$

$$W_2 = 18 \text{ g}$$

14. $$i = \frac{\text{observed property}}{\text{Theoritical property (calculated)}}$$

Given ΔT_f = 0.680 K

m = 0.2 m

ΔT_f (observed) = 0.680 K

ΔT_f (calculated) = K_f . m

= 1.86 K Kg mol^{-1} × 0.2 mol Kg^{-1}

= 0.372 K

$$i = \frac{(\Delta T_f) \text{ observed}}{(\Delta T_f) \text{ calculated}} = \frac{0.680 \text{ K}}{0.372 \text{ K}} = 1.82$$

MCQ

1. $$\text{molality} = \frac{\text{number of moles of solute}}{\text{weight of solvent (in kg)}}$$

$$= \frac{\left(\frac{1.8}{180}\right)}{0.25} = \frac{0.01}{0.25} = 0.04 \text{ M}$$

option (d) is correct.

2. option (d) is correct. Molality and mole fraction are independent of temperature.

3. $M_1 \times V_1 = M_2 \times V_2$ [$\because$ 0.1 M $Al(OH)_3$ gives 3 × 0.1 = 0.3 M OH^- ions]

$0.3 \times V_1 = 0.1 \times 21$

$$V_1 = \frac{0.1 \times 21}{0.3} = 7 \text{ ml}$$